PINCHED JORDAN CURVES, CANNON-THURSTON MAPS, AND THEIR DECOMPOSITIONS

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Abstract. TODO

Contents

1. Introduction 1
2. Definitions and Preliminaries 1
3. Pinched Jordan Curves and Inner Diameter Distance 5
4. The Inner Decompositions of a Pinched Jordan Curve 10
5. Special Decompositions are Dendrites 13
6. Self Bumping and Laminations 14
Appendix A. Cannon-Thurston Maps and Pseudo-Isotopies 14
References 16

1. Introduction

Definition 1.1. A *pinched Jordan curve* is a map $\varphi_\infty : S^1 \to S^2$ and a choice of pseudo-isotopy φ_s, $s \in [0, \infty]$, landing on φ_∞. To avoid unnecessary quotients, we require each preimage $\varphi_s^{-1}(p)$ is totally disconnected. ♮

Remark 1.2. Given a pseudo-isotopy $\varphi : S^1 \times I \to S^2$ where $\varphi_\infty(S^1)$ does not have totally disconnected points preimages, we may pass to a quotient of S^1 to obtain a new map $\hat{\varphi}_\infty$ which has totally disconnected preimages. Adjusting the approximations is similarly easy: any arcs collapsed by φ_∞ must get uniformly small under the φ_s, so we may modify each φ_s in a neighborhood of each collapsed interval to produce embeddings $\hat{\varphi}_s$ such that $\hat{\varphi}_s \to \hat{\varphi}_\infty$. ♮

Remark 1.3. While our definition requires a *specific* pseudo-isotopy, it is a consequence of Theorem 6.1 that our results depend on an isotopy class of pseudo-isotopies landing on φ_∞; see Section 6. ♮

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2. Definitions and Preliminaries

In this section, we give the topological background required for the rest of the paper. We cover Hausdorff convergence, upper semi-continuous decompositions, and locally flat isotopies.

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2.1. **Convergence of Sets.**

2.1.1. *Hausdorff Convergence.* Let X be a metric space. Writing $N_\varepsilon(A) = \bigcup_{a\in A} B_\varepsilon(a)$, we may endow $\mathcal{P}(X)$, the power set of X, with the *Hausdorff distance*:

$$d_H(A,B) = \inf\{r \mid A \subset N_r(B) \text{ and } B \subset N_r(A)\}$$

One need not work very hard to construct examples showing that both inclusions are needed for d_H to be symmetric. When X is a compact metric space, then d_H gives a metric on the closed subsets of X—d_H must be restricted the set of closed subsets of X rather that all of $\mathcal{P}(X)$ as $d_H(A,\overline{A}) = 0$ for all sets A. The Hausdorff distance sees two sets as close if they they are everywhere point-wise close. Indeed, we will often establish that two sets are Hausdorff-close via the following characterization:

$$d_H(A,B) < \varepsilon \Leftrightarrow A \subset N_\varepsilon(B) \text{ and } B \subset N_\varepsilon(A)$$

We write $\lim_H(A_n) = A$ or $A_n \xrightarrow{HD} A$ if A is closed and $d_H(A_n, A) \to 0$ as $n \to \infty$.

Lemma 2.1. *Hausdorff convergence respects uniformly continuous maps and uniformly convergent sequences. Let $f_k : X \to Y$ be a sequence of maps between compact metric spaces such that $f_k \to f_\infty$ uniformly. If $A_k \xrightarrow{HD} A$ then $f_k(A_k) \xrightarrow{HD} f_\infty(A)$.*

Proof. Fix $\varepsilon > 0$. As $f_k \to f_\infty$ is uniformly convergent, Arzelà–Ascoli implies that the family $\{f_k\}_{k\in\mathbb{Z}^+\cup\{\infty\}}$ is uniformly equicontinuous. As such, there is $\delta > 0$ such that if $d_X(x,y) < \delta$ then $d_Y(f_k(x), f_k(y)) < \varepsilon$ for any $k \in \mathbb{Z}^+ \cup \{\infty\}$. Further, there is N_1 such that if $k > N_1$, then $d_Y(f_k(x), f_\infty(x)) < \varepsilon$. As $A_k \xrightarrow{HD} A$, there is some $N_2 > 0$ such that if $k > N_2$, then $d_{HD}(A_k, A) < \delta$.

Now suppose that $k > \max\{N_1, N_2\}$. Pick $y \in f(A)$ and $x \in f^{-1}(y)$. As $d_{HD}(A_k, A) < \varepsilon$, there is some $x_k \in A_k$ such that $d_X(x, x_k) < \delta$. Then

$$d_Y(y, A_k) \le d_Y(y, f_k(x_k)) \le d_Y(f(x), f_k(x)) + d_Y(f_k(x), f_k(x_k)) \le 2\varepsilon,$$

hence $f(A) \subset N_{2\varepsilon}(f_k(A_k))$.

Now suppose that $y \in f_k(A_k)$. Select $x \in f_k^{-1}(y)$. As $x \in A_k$ there is some $x' \in A$ such that $d_X(x, x') < \delta$. Then

$$d_Y(y, f(A)) \le d_Y(f_k(x), f(x')) \le d_Y(f_k(x), f_k(x')) + d_Y(f_k(x'), f(x')) \le 2\varepsilon.$$

We conclude that $f_k(A_k) \subset N_{2\varepsilon}(f(A))$. We have shown that if k is sufficiently large, then $d_{HD}(f_k(A_k), f(A)) < 2\varepsilon$. □

We will occasionally use a slightly different formulation for convergence. Suppose that X is a compact metric space and let X_n be a sequence of closed subsets of X. We define the limits inferior and superior as follows:[1]

$$\liminf X_n = \{x \in X \mid \text{ Every neighborhood of } x \text{ intersects all but finitely many } X_n\}$$
$$\limsup X_n = \{x \in X \mid \text{ Every neighborhood of } x \text{ intersects infinitely many } X_n\}$$

Equivalently, $x \in \liminf X_n$ if one can find a sequence $x_n \in X_n$ such that $x_n \to x$ and $x \in \limsup X_n$ if one can find a subsequence n_k and a sequence $x_k \in X_{n_k}$ such that $x_k \to x$. It is immediate that $\liminf X_n \subset \limsup X_n$. For any sequence of sets X_n, $\limsup X_n$ and $\liminf X_n$ are closed. Indeed, if $x_n \in \liminf X_n$ is a sequence converging to some $x \in X$

[1]The reader may recognize these as *Kuratowski* convergence, a formulation for taking limits of sets in general topological spaces. In a compact metric space these formulations agree so we do not make the distinction. See [Hocking Young Sec ??].

then any neighborhood U of x is also a neighborhood of x_n for some n. As $x_n \in \liminf X_n$, all but finitely many of the X_n meet U, hence $x \in \liminf X_n$—so $\liminf X_n$ is closed. The same argument applies to $\limsup X_n$.

Lemma 2.2. *Suppose that X_n is a sequence of closed sets in a compact metric space X. If $Y = \liminf X_n = \limsup X_n$, then $X_n \xrightarrow{HD} Y$.*

Proof. Fix $\varepsilon > 0$. First, we will establish that if n is sufficently large that $X_n \subset N_\varepsilon(Y)$. Suppose to the contrary, that is, there is a subsequence n_k such that for all k $X_{N_k} \setminus Y \neq \varnothing$; select an $x_k \in X_{N_k} \setminus Y$ for each k. After passing to a subsequence, we may assume that $x_k \to x$. Each x_k satisfies $d(x_k, Y) > \varepsilon$, so $d(x, Y) \geq \varepsilon$, but $x \in \limsup X_n = Y$, a contradiction. Thus, for all but finitely many n, $X_n \subset N_\varepsilon(Y)$.

Now we will establish that for large n, $Y \subset N_\varepsilon(X_n)$. Supposing otherwise, we obtain a subsequence n_k such that $Y \setminus N_\varepsilon(X_{n_k})$ is nonempty. For each k select some $x_k \in Y \setminus N_\varepsilon(X_{n_k})$; up to subsequence $x_k \to x \in Y$. We now claim that $N_{\varepsilon/2} \cap X_{n_k}$ is empty for all large k. Indeed if k is large enough that $d(x_k, x) < \varepsilon/2$, then

$$d(x_k, X_{n_k}) \leq d(x_k, x) + d(x, X_{n_k}) \leq \varepsilon.$$

This contradicts the fact that $x_k \notin N_\varepsilon(X_{n_k})$ so we conclude that for large k $N_{\varepsilon/2}(x) \cap X_{n_k} = \varnothing$. On the other hand, $x \in \liminf X_n$ so $N_{\varepsilon/2}(x)$ must contain all but finitely many of the X_n. This contradiction establishes that, for large n, $Y \subset N_\varepsilon(X_n)$. Thus, $X_n \xrightarrow{HD} Y$. □

We end this section with a number of useful facts concerning Hausdorff convergence.

Proposition 2.3 ([HY88], Theorem 2-101)**.** *If X_n is a sequence of connected sets in a compact Hausdorff space X and if $\liminf X_n$ is nonempty, then $\limsup X_n$ is connected.*

Corollary 2.4. *Suppose $X_n \to X$ is a convergent sequence in a compact Hausdorff space. If each X_n is connected, so is X.*

Proposition 2.5 ([HY88], Theorem 2-102)**.** *If M is a compact metric space, then every sequence of subsets of M has a Hausdorff convergent subsequence.*

Fact 2.6. *If $A_n \xrightarrow{HD} A$, then $\operatorname{diam} A_n \to \operatorname{diam} A$*

Fact 2.7. $\operatorname{diam}(A \cup B) \leq \operatorname{diam} A + \operatorname{diam} B + d(A, B)$, *where* $d(A, B) = \inf\{d(a, b) \mid a \in A, b \in B\}$.

2.2. **Decompositions.** For a space X, a *decomposition* of X, $\mathcal{D}$, is a partition of X into closed subsets. That is, we can write $X = \bigsqcup_{D \in \mathcal{D}} D$ and each $D \subset X$ is closed. For $x \in X$, we will write D_x for the unique decomposition element containing x. This gives a set function $\pi : X \to \mathcal{D}$ given by $x \mapsto D_x$. We may endow $\mathcal{D}$ with topology by declaring that $\mathcal{A} \subset \mathcal{D}$ is open when $\pi^{-1}(\mathcal{A})$ is open in X. This is precisely the quotient topology inherited from X—so π is continuous. In light of this, by a slight abuse of notation we will usually use $\mathcal{D}$ to represent both subsets of X and for the quotient space $X/\mathcal{D}$. If we wish to stress the fact that we are treating $\mathcal{D}$ as a space on its own, we will refer to it as a *decomposition space*.

Given any other map $f : X \to Y$ which is constant on $\mathcal{D}$, we obtain a unique $g : \mathcal{D} \to Y$ such that $f = g \circ \pi$. We say that a set $W \subset X$ is *$\mathcal{D}$-saturated* if W can be written as a union of decomposition elements. If $D \in \mathcal{D}$ contains more than one point, we say that D is a *non-degenerate* element of $\mathcal{D}$. Otherwise D is called a *degenerate* decomposition element. We refer the curious reader to [Dav] for a full treatment of the theory of decompositions of manifolds and [Kal21] for a slightly more general treatment.

Definition 2.8. A decomposition $\mathcal{D}$ of a space X is *upper semi-continuous* if, given any open set $U \subset X$ and some $D \subset U$, there is an open set $V \subset U$ containing D such that every decomposition element intersecting V is contained in U. ♮

Upper semi-continuous decompositions are meant to generalize point-preimages of a surjective map $f : X \to Y$ from a compact space to a Hausdorff space one. We create the decomposition $\mathcal{D}$ by dividing X into point-preimages:

$$\mathcal{D} = \{f^{-1}(y) \mid y \in Y\} \setminus \varnothing.$$

$\mathcal{D}$ is then upper semicontinuous. It is a classic result that Y and $\mathcal{D}$ are therefore homeomorphic. While the point-preimage example is the model for upper semi-continuous decompositions, there are a number of other useful characterizations coming from a combination of [Kal21, Lemma 2.7] and [FL25, Lemma 2.8]

Proposition 2.9. *Let $\mathcal{D}$ be a decomposition of a compact metric space X. The following are equivalent:*

(1) $\mathcal{D}$ is upper semi-continuous
(2) For every closed $K \subset X$, then set $\bigcup\{D \in \mathcal{D} \mid D \cap K \neq \varnothing\}$ is closed in X.
(3) Let A_i be a sequence of decomposition elements with $\lim A_i = A$. Then there is a $D \in \mathcal{D}$ with $A \subset D$.
(4) The quotient $\pi : X \to \mathcal{D}$ is closed.
(5) As a decomposition space, $\mathcal{D}$ is Hausdorff.

2.3. **Embeddings and Isotopy Extension.**

Definition 2.10. A pseudo-isotopy is a continuous family of maps $f_t : X \to Y$, $t \in [0, 1]$ such that for all $t < 1$, f_t is an embedding. ♮

The map $f_1 : X \to Y$ may or may not be an embedding. If $J \subset [0, 1)$ is a closed interval, then the collection $f_t, t \in J$ is an *isotopy*.

Definition 2.11. An embedding of topological manifolds $N^n \subset M^m$ is called *locally flat* if for every $p \in N$, there is a chart $\phi_p : \mathbb{R}^m \to M$ centered at p such that $\mathbb{R}^{m-n} \subset \mathbb{R}^m$ is mapped to N. Embeddings which are not locally flat are called *wild* embeddings. ♮

Locally flat embeddings serve as the generalization of smoothly embedded submanifolds; wild embeddings can introduce pathologies. See, for example, [CITE: Rolfsen? Other?]. An embedding of spheres $S^n \hookrightarrow S^m$ is called *flat* if S^n be realized as the image of the equatorial n-sphere under a homeomorphism $S^m \to S^m$. The famous Schoenflies Theorem [CITE: Hocking Young? Earlier?] states that every S^1 embedded in S^2 is flat.

Definition 2.12 ([EK71])**.** An isotopy of manifolds $f_t : N^n \to M^m$, $t \in I$, is *locally flat* if for every $(p, t) \in N \times I$, there is a neighborhood $[t_0, t_1]$ of $t \in I$ and level-preserving embeddings $\alpha : \mathbb{R}^n \times [t_0, t_1] \to N \times I$ and $\beta : \mathbb{R}^n \times \mathbb{R}^{m-n} \times [t_0, t_1] \to M \times I$ onto neighborhoods of (p, t) and $(f_t(p), t)$ respectively such that the following diagram commutes.

$$\begin{array}{ccc} \mathbb{R}^n \times 0 \times [t_0, t_1] & \hookrightarrow & \mathbb{R}^n \times \mathbb{R}^{m-n} \times [t_0, t_1] \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} \\ N \times I & \xrightarrow[(p,t)\mapsto(f_t(p),t)]{} & M \times I \end{array}$$

In particular, this means that if U is a sufficiently small locally flat chart around $f_t(p)$ for some $p \in N$, then the isotopy locally extends to $F : U \times [t_0, t_1] \to M$ such that for all $t' \in [t_0, t_1]$, $F_{t'}(U)$ is a locally flat chart for $f_{t'}(p)$. ♮

As long as N is a locally flat proper submanifolds, a locally flat isotopy may be viewed as a level preserving map $N \times I \to M \times I, (p, t) \mapsto (f_t(p), t)$ which locally flat in the usually sense.

Definition 2.13. An isotopy $f_t : X \to Y$, $t \in I$, is *realized by an ambient isotopy* if there is an isotopy $F_t : Y \to Y$, $t \in I$, which cover the f_t in the sense that $F_1 = \mathbb{1}_Y$ and for all $t \in I$, $f_t = F_t f_0$. ♮

Theorem 2.14 ([EK71], Cor. 1.4)**.** *Suppose $f_t : N \to M$, $t \in I$ is a locally flat proper isotopy of a compact manifold N into a manifold M. Then f_t is realized by an ambient isotopy of M.*

In the context of Schoenflies theorem, one wonders which *isotopies* $f_t : S^n \subset S^m$ are locally flat. Work by Oversteegen and Tymchatyn [OT08] answers this in the case where $n = 1$ and $m = 2$:

Theorem 2.15 ([OT08], Thm. 7.2)**.** *Suppose that $Z \subset \mathbb{R}^2$ is compact and connected and that $f_t : Z \to S^2$ is an isotopy such that $f_0 = \mathbb{1}_Z$. Then there exists an ambient isotopy $F_t : S^2 \to S^2$ extending f_t.*

3. Pinched Jordan Curves and Inner Diameter Distance

In this section we introduce family of metrics d_s^+, $s \in [0, \infty]$, associated to a pinched Jordan curve which contains information about its pseudo-isotopy. We state the definition of a pinched Jordan curve again for convenience:

Definition 1.1. A *pinched Jordan curve* is a map $\varphi_\infty : S^1 \to S^2$ and a choice of pseudo-isotopy φ_s, $s \in [0, \infty]$, landing on φ_∞. To avoid unnecessary quotients, we require each preimage $\varphi_s^{-1}(p)$ is totally disconnected. ♮

Notation. We fix the following notation which will be used throughout: For $s \in [0, \infty]$, let $C_s = \varphi_s(S^1)$. For $s < \infty$, C_s is a Jordan curve; thus, given some fixed orientation on S^1, C_s divides S^2 into two disks. Let $\mathring{C}_s^+$ and $\overline{C_s^+}$ denote the open and closed disks to the left of C_s. Likewise, we write $\mathring{C}_s^-$, and $\overline{C_s^-}$ for the open and closed disks to the right of C_s. ♮

Definition 3.1. Suppose φ_s, $s \in [0, \infty]$, is a pinched Jordan curve and let $p \in S^2 \setminus C_\infty$. We say that p is *to the left of* (resp. *to the right of*) C_∞ if the winding number of φ_∞ around p is 1 (resp. -1). ♮

Lemma 3.2. *Suppose that φ_s, $s \in [0, \infty]$, is a pinched Jordan curve. Then,*

1. *Every $p \in S^2 \setminus C_\infty$ is either to the left or the right of C_∞*
2. *If p and q lie in the same component of $S^2 \setminus C_\infty$, then p is to the left of C_∞ if and only if q is.*

Proof. Let U be a component of $S^2 \setminus C_\infty$ and suppose $p \in U$. We may select a large s_0 such that such that $p \notin C_s$ for all $s \in [s_0, \infty]$. As C_{s_0} is a Jordan curve, its winding number around p is ± 1; as the winding number is invariant under a homotopy missing p, the winding number of C_∞ around p is also ± 1. If $q \in U$, let α denote an arc in U with endpoints p and q. Picking a larger s_0 if necessary, we may assume that $\alpha \cap C_s = \varnothing$ for all $s \in [s_0, \infty]$. Thus, C_{s_0} has the same winding number around any point in α; homotopy invariance once again implies C_∞ has the same winding number around p and q. □

Definition 3.3. We say that a component U of $S^2 \setminus C_\infty$ is *to the left* of C_∞ if some (any) $p \in U$ is to left of C_∞. We say that a pinched Jordan curve C_∞ has *no left side* if every complementary region is to the right of C_∞. We similarly define components to the right of C_∞ and pinched Jordan curves with no right side. ♮

The following definition is inspired by [Cal07, Construction 10.11] and by section 10.4.1 more broadly.

Definition 3.4. Suppose φ_s, $s \in [0,\infty]$ is a pinched Jordan curve. Given $x, y \in S^1$ and $s < \infty$, define the *s-inner diameter* between x and y to be

$$d_s^+(x,y) = \inf\{\operatorname{diam} X \mid X \subset \overline{C_s^+} \text{ is connected, closed, and } \varphi_s(x), \varphi_s(y) \in X\}$$

The sets X above are called *s-stems* for x and y (or just a *stem* when x, y, and s are understood). ♮

Our goal this section is to establish the following description of d_s^+:

Theorem 3.5. *Suppose that φ_s, $s \in [0,\infty]$, is a pinched Jordan curve and let d_s^+ the s-inner diameter function. Then*

1. *d_s^+ is a metric on S^1 for all $s < \infty$.*
2. *The family $\{d_s^+ \mid s < \infty\}$ is continuous as function $S^1 \times S^1 \times [0,\infty) \to \mathbb{R}$.*
3. *There is a map $d_\infty^+ : S^1 \times S^1 \to \mathbb{R}$ which is a pseudo-metric on S^1; it is the unique continuous extension of the family $\{d_s^+\}$ to a function $S^1 \times S^1 \times [0,\infty] \to \mathbb{R}$.*

Remark 3.6. As discussed in the introduction, we are mainly interested in pinched Jordan curves where $\varphi_\infty : S^1 \to S^2$ is surjective—meaning there is no left or right side. In this section, we describe d_s^+ and its properties, but one may similarly define d_s^- with stems in $\overline{C_s^-}$. In the sequel we will use both families, but at the present we need only consider d_s^+. ♮

Definition 3.7. Suppose $\varphi_s, s \in [0,\infty]$, is a pinched Jordan curve and X is an s-stem for $x, y \in S^1$. We say that X is *taut* if $\operatorname{diam} X = d_s^+(x,y)$. We say that X is *proper* if is represented by a proper arc in $\overline{C_s^+}$. ♮

Lemma 3.8 (Existence of Taut and Proper Stems)**.** *Suppose $\varphi_s, s \in [0,\infty]$ is a pinched Jordan curve. Given $x, y \in S^1$ and $s < \infty$:*

1. *There is an s-stem X such that $\operatorname{diam} X = d_s^+(x,y)$*
2. *Given $\varepsilon > 0$ and any s-stem X for x and y, there is a proper arc $\gamma \subset \overline{C_s^+}$ which is also an s-stem for x and y such that $\operatorname{diam}\gamma \le \operatorname{diam} X + \varepsilon$.*

Proof.

1. Let X_n be a sequence of closed, connected sets containing $\varphi_s(x)$ and $\varphi_s(y)$ such that $\operatorname{diam} X_n \to d_s^+(x,y)$. Passing to a subsequence if necessary, assume $X_n \xrightarrow{HD} X$. As $\varphi_s(x), \varphi_s(y) \in X_n$, for all n, $\varphi_s(x), \varphi_s(y) \in X$. By Corollary 2.4, X is connected. Finally, $d_s^+(x,y) = \lim_n \operatorname{diam} X_n = \operatorname{diam} \lim_H X_n = \operatorname{diam} X$, as required.
2. Consider $N_\varepsilon(X)$, the ε-neighborhood of X in $\overline{C_s^+}$. $N_\varepsilon(X)$ is path connected, contains $\varphi_s(x)$ and $\varphi_s(y)$, and satisfies $\operatorname{diam} N_\varepsilon(X) \le \operatorname{diam} X + \varepsilon$—thus, we select some proper arc $\gamma \subset N_\varepsilon(X) \subset \overline{C_s^+}$ containing $\varphi_s(x)$ and $\varphi_s(y)$.

□

Proposition 3.9. *For each $s < \infty$, d_s^+ is a metric on S^1.*

Proof. Symmetry and positive definiteness follow directly from definition and the fact that each φ_s is an embedding. For the triangle inequality, let X be a taut stem for $x, y \in S^1$ and let Y be a taut stem for $y, z \in S^1$. Then $X \cup Y$ is an s-stem for x and z, hence

$$d_s^+(x,z) \leq \operatorname{diam}(X \cup Y) \leq \operatorname{diam} X + \operatorname{diam} Y = d_s^+(x,y) + d_s^+(y,z),$$

where the final inequality follows from the fact that X and Y intersect at $\varphi_s(y)$.† □

lk: And an implicit use of Fact 2.7. Call this out directly?

3.1. **Curve Pushing and Continuity of Inner Distance.** While we cannot say much about C_∞ for a generic pinched Jordan curve, we may still define d_∞^+ as the limit of the d_s^+. The goal of this section is two-fold: first, we wish to show that $\lim_s d_s^+(x,y)$ exists for any pair $x, y \in S^1$. Second, we use this fact to show d_s^+ is continuous as a function $[0,\infty] \times S^1 \times S^1 \to \mathbb{R}$. The main tool will be the following result:

Proposition 3.10 (Stem pushing)**.** *Suppose $\varphi_s : S^1 \to S^2$, $s \in [0,1]$, is an isotopy and $\alpha : I \to S^2$ is a proper arc in $\overline{C_0^+}$ between $\varphi_0(x)$ and $\varphi_0(y)$. Suppose further that for all $s \in [0,1]$, $d_{\sup}(\varphi_s, \varphi_0) < \varepsilon$. There is a 1-stem $Y \subset \overline{C_1^+}$ for x and y such that* $\operatorname{diam} Y \leq \operatorname{diam} \alpha(I) + 4\varepsilon$

Essentially, our proof is as follows: Suppose that $\varphi_s : S^1 \to S^2$, $s \in [0,1]$, is an isotopy that moves points at most ε. Given α_0, a proper 0-stem for x and y, we want to produce a 1-stem Y for x and y such that $Y \subset N_{2\varepsilon}(\alpha_0)$. To do so, whenever the isotopy would cross α_0, we want the isotopy φ_s to instead "push" α_0 along with it. As such, we will keep track of the set $\{z \in S^1 \mid \varphi_s(z) \in \alpha_0, \text{ for some } s \in [0,1]\}$—in the settings of Lemmas 3.13 and 3.14 this set has slightly different definitions, but we name this set K in both circumstances. We will define Y as a union of (a subset of) $\alpha_0 \cap \overline{C_1^+}$ and (a subset of) $\varphi_1(K)$.

Proposition 3.9 establishes property 1 of Theorem 3.5, so it remains to show properties 2 and 3. After some technical preliminaries in Lemmas 3.11 and 3.12, we do the bulk of work proving Proposition 3.10 in Lemmas 3.13 and 3.14. We then finish the proof of Theorem 3.5. For the rest of the section, we adopt the following convections: let $Q = I \times I$ and write T, B, L, R for the top, bottom, left, and right edges of Q.

3.1.1. *Technical Lemmas.* Our first technical lemma guarantees that our extension satisfies $d_\infty(x,x) = 0$ and will be used in the sequel to handle sequences $x_n \in S^1$ associated to a sequence of times $s_n \to \infty$. Our second lemma concerns a descriptions of closed sets of Q which separate L and R.

Lemma 3.11. *Suppose $x_n \to x \in S^1$ and $s_n \to \infty$. Then $d_{s_n}(x_n, x) \to 0$.*

Proof. Let I denote a small interval around $x \in S^1$. Passing to the tail of the sequence, we any assume that $x_n \in I$ for all n. Let I_n denote the component of $S^1 \setminus \{x, x_n\}$ contained in I. As $\varphi_{s_n}(I_n)$ is an s_n-stem for x and y we know $d_{s_n}^+(x,y) \leq \operatorname{diam} \varphi_{s_n}(I_n)$. Further, $I_n \xrightarrow{HD} \{x\}$, hence $\varphi_{s_n}(I_n) \xrightarrow{HD} \{\varphi_\infty(x)\}$ by Lemma 2.1. Taking diameters, $d_{s_n}^+(x,y) \leq \operatorname{diam} \varphi_{s_n}(I_n) \to 0$. □

Lemma 3.12. *Suppose $X \subset Q$ is closed and that both $X \cap T$ and $X \cap B$ are nonempty. If no component Y of X meets both T and B, then there is a path $\alpha : [0,1] \to I \times I \setminus X$ such that $\alpha(0) \in L$ and $\alpha(1) \in R$.*

Proof. As X has at least two component, partition X into $X = X_T \sqcup X_B$ where $T \subset X_T$ and $B \subset X_B$ and $X_T \cap X_B = \varnothing$. Embed Q as the square in $\mathbb{R}^2$ with corners $(0,0), (0,1), (1,0), (1,1)$. We extend the sets X_T and X_B as $Y_T = X_T \cup \mathbb{R} \times [1,\infty)$ and

$Y_B = X_B \cup \mathbb{R} \times (-\infty, 0]$; note that $Y_T \cap Y_B = \varnothing$ as well. Let U_T and U_B denote the complements of Y_T and Y_B respectively. We apply the Mayer-Vietoris sequence to $\mathbb{R}^2 = U_T \cup U_B$:

$$0 \to \tilde{H}_0(U_T \cap U_B) \xrightarrow{\psi} \tilde{H}_0(U_T) \oplus \tilde{H}_0(U_B) \to 0$$

is exact.

Pick points $p_L \in L \setminus X$ and $p_R \in R \setminus X$. As $U_B \setminus Q$ is path connected, we may join p_L and p_R with a path $\gamma_B \subset U_B$ which travels around and above Q—thus p_L and p_R are homologous as elements of $\tilde{H}_0(U_B)$. Likewise, we may connect them with a similar path $\gamma_T \subset U_T$ so they are also homologous in $\tilde{H}_0(U_T)$. Now consider $[p_L], [p_R] \in \tilde{H}_0(U_T \cap U_B)$; as $[p_L] = [p_R]$ in both $\tilde{H}_0(U_T)$ and $\tilde{H}_0(U_B)$, $\psi([p_L] - [p_R]) = 0$, but ψ is an isomorphism hence $[p_L]$ and $[p_R]$ are homologus in $\tilde{H}_0(U_T \cap U_B)$.

This produces an arc γ such that $\gamma(0) = p_L$, $\gamma(1) = p_R$ and $\gamma(t) \in U_T \cap U_B$ for all t. It may not be the case that $\gamma_t \in Q$ for all t, but some component of $\gamma(I) \cap Q$ must hit both the left and right edges—we take α to be this component. □

3.1.2. *Pushing Stems.* For the next two lemmas, we adopt the following conventions: We write E for the equator of S^2 and $\mathring{H}_\pm, \overline{H_\pm}$ for the open and closed northern and southern hemispheres respectively. In order to push around stems, we need a change of perspective; instead of the stem α being fixed while $\varphi_s(S^1)$ moves, we deal with a fixed circle (the equator) and an isotopy of arcs starting in the northern hemisphere. Lemma 3.13 handles the perspective of a fixed equator and Lemma 3.14 translates between the two perspectives.

Lemma 3.13. *Fix $x, y \in E \subset S^2$, points in the equator, and let γ_s, $s \in [0,1]$, be a continuous family of arcs in S^2 such that*

(1) For all s, $\gamma_s(0) = x$ and $\gamma_s(1) = y$.
(2) γ_0 is a proper arc in $\overline{H_+}$

Let $K = \{z \in E \mid z = \gamma_s(t), \text{ for some } s,t \in [0,1]\}$. Then there is a connected set $Y \subset K \cup \left(\gamma_1(I) \cap \overline{H_+}\right)$ containing x and y.

Proof. Define $\Gamma : Q \to S^2$ by $\Gamma(s,t) = \gamma_s(t)$. Let $A = \Gamma^{-1}(E)$ and $B = \Gamma^{-1}(\overline{H_+}) \cap R$. Note that $(s,t) \in A$ if and only if $\Gamma(s,t) = \gamma_s(t) \in E$—so in this circumstance $\Gamma(s,t) \in K$. Similarly, if $p \in B$, then $\Gamma(p) \in \gamma_1(I) \cap \overline{H_+}$ by construction.

Claim. There does not exist a path $\alpha : I \to Q \setminus (A \cup B)$ such that $\alpha(0) \in L$ and $\alpha(1) \in R$. ↯

Proof (of claim). Suppose otherwise. As $\alpha(I) \cap A = \varnothing$, $\Gamma \circ \alpha$ is a path in $S^2 \setminus E$. We know that $\alpha(0) \in \gamma_0 \setminus E$, hence $\alpha(0) \in \mathring{H}_+$. On the other hand, $\alpha(1) \in \gamma_1 \setminus H_+$, hence $\alpha(1) \in S^2 \setminus \mathring{H}_-$. But no arc can cross from $\mathring{H}_+$ to $\mathring{H}_-$ without intersecting E, a contradiction. □

As $L, R \subset A \cup B$, $A \cup B$ is closed, and no such arc α exists, by the contrapositive of Lemma 3.12, some component X of $A \cup B$ contains both L and R. We set $Y = \Gamma(X)$. We have seen that $\Gamma(A) \subset K$ and $\Gamma(B) \subset \gamma_1(I) \cap \overline{H_+}$, hence $Y \subset K \cup \left(\gamma_1 \cap \overline{H_+}\right)$, as required. □

FIGURE – the isotopy of an arc in H_+, the square $I \times I$, and the connected sets we turn into the stems.

Lemma 3.14. *Let $\varphi_s : S^1 \to S^2$, $s \in [0,1]$, be an isotopy and $\alpha : I \to S^2$ be a proper path in $\overline{C_0^+}$ such that $\alpha(0) = \varphi_0(x)$ and $\alpha(1) = \varphi_0(y)$. Let $\beta_s = \varphi_{[0,s]}(x) * \alpha * \varphi_{[0,s]}(y)$, where $*$ is concatenation and we reverse terminal path, and define $K = \{\varphi_1(z) \mid \varphi_s(z) = \beta_s(t) \text{ for some } s,t \in [0,1]\}$. There is a connected set $Y \subset K \cup (\alpha(I) \cap \overline{C_1^+})$ containing $\varphi_1(x)$ and $\varphi_1(y)$.*

Proof. Identifying S^1 with $E \subset S^2$ and using Theorem 2.15, extend φ_s to a family of maps $\Phi_s : S^2 \to S^2$. Let β_s be as above; then $\gamma_s = \Phi_s^{-1} \circ \beta_s$ is a isotopy of paths in S^2 with fixed endpoints and $\gamma_0((0,1)) \subset \mathring{H}_+$.

Let $K_\Phi = \{z \in E \mid z = \gamma_s(t) \text{ for some } s,t \in [0,1]\}$ and notice $K = \Phi_1(K_\Phi)$. Applying Lemma 3.13, we obtain a connected set $\tilde{Y} \subset K_\Phi \cup (\gamma_1(I) \cap H_+)$. We claim that $Y = \Phi_1(\tilde{Y})$ is the required set. By construction,

$$\begin{aligned} Y &\subset \Phi_1(K_\Phi) \cup \Phi_1(\gamma_1(I) \cap H_+) \\ &\subset \Phi_1(K_\Phi) \cup (\Phi_1\Phi_1^{-1}\beta_1(I) \cap \Phi_1(H_+)) \\ &= K \cup (\beta_1(I) \cap \overline{C_1^+}). \end{aligned}$$

□

We are now in a position to prove Proposition 3.10, which we restate here for convenience:

Proposition 3.10 (Stem pushing)**.** *Suppose $\varphi_s : S^1 \to S^2$, $s \in [0,1]$, is an isotopy and $\alpha : I \to S^2$ is a proper arc in $\overline{C_0^+}$ between $\varphi_0(x)$ and $\varphi_0(y)$. Suppose further that for all $s \in [0,1]$, $d_{\sup}(\varphi_s, \varphi_0) < \varepsilon$. There is a 1-stem $Y \subset \overline{C_1^+}$ for x and y such that* $\operatorname{diam} Y \leq \operatorname{diam} \alpha(I) + 4\varepsilon$

Proof. We produce the set Y using Lemma 3.14. Y is a 1-stem for x and y by construction, so all that remains is to show diameter bound. As in the statement of Lemma 3.14, let $\beta_s = \varphi_{[0,s]}(x) * \alpha * \varphi_{[0,s]}(y)$ and write $K = \{\varphi_1(z) \mid \varphi_s(z) = \beta_s(t) \text{ for some } s,t \in [0,1]\}$. Suppose $p \in Y$; we will show that $d_{S^2}(p, \alpha(I)) < 2\varepsilon$. As $d_{\sup}(\varphi_s, \varphi_0) < \varepsilon$, $d_{HD}(\beta_s(I), \alpha(I)) < \varepsilon$. Thus, $\operatorname{diam} \beta_s(I) \leq \operatorname{diam} \alpha(I) + 2\varepsilon$ and therefore if $p \in \beta_s(I)$ then $d_{S^2}(p, \alpha(I)) < 2\varepsilon$.

Now suppose that $p \in K$. By definition, $p = \varphi_1(z)$ and there are $s_0, t_0 \in [0,1]$ such that $\varphi_{s_0}(z) = \beta_{s_0}(t_0)$. But then

$$d_{S^2}(p, \alpha(I)) \leq d_{S^2}(p, \varphi_{s_0}(z)) + d_{S^2}(\varphi_{s_0}(z), \alpha(I))$$

The first term is less than ε by assumption, and the second is less that ε as $\varphi_{s_0}(z) \in \beta_{s_0}(I)$ and $d_{HD}(\beta_{s_0}(I), \alpha(I)) < \varepsilon$ as above. As $Y \subset K \cup \beta_1$, $d_{S^2}(p, \alpha(I)) < 2\varepsilon$ and the result follows. □

3.1.3. *Continuity and Continuous Extension.* With the main technical tool established, we are finally in a position to finish the proof of Theorem 3.5.

Proposition 3.15. *Suppose φ_s is a pinched Jordan curve and let $s_n \to s \in [0,\infty]$ be any convergent sequence. For any $x, y \in S^1$, $\lim_{n\to\infty} d^+_{s_n}(x,y)$ exists. If $s < \infty$, then* $\lim_{n\to\infty} d^+_{s_n}(x,y) = d^+_s(x,y)$.

Proof. To show that the limit exists, we will show that $d^+_{s_n}(x,y)$ is Cauchy. Fix $\varepsilon > 0$. As $\varphi_{s_n} \to \varphi_s$ uniformly, we may pass to a tail of the sequence and guarantee that $d_{\sup}(\varphi_{s_n}, \varphi_{s_m}) < \varepsilon$ for all m, n. Let X be a proper s_n stem such that $\operatorname{diam} X \leq d_{s_n}(x,y) + \varepsilon$ and write $\tilde{X}$ the s_m-stem obtained by applying Proposition 3.10 to X. Then,

$$d^+_{s_m}(x,y) \leq \operatorname{diam} \tilde{X} \leq \operatorname{diam} X + 4\varepsilon \leq d^+_{s_n}(x,y) + 5\varepsilon.$$

Likewise, $d^+_{s_n}(x,y) \leq d^+_{s_m}(x,y) + 5\varepsilon$ by a symmetric argument; whence $\lim_{s_n \to s} d^+_{s_n}(x,y)$ exists.

Now suppose that $s < \infty$ and X_n is a taut s_n-stem. Passing to a subsequence if necessary, we may assume $X_n \xrightarrow{HD} X$. Evidently, X is an s-stem so $d^+_s(x,y) \leq \lim_{n\to\infty} d^+_{s_n}(x,y)$. Now

let Y be an ε-efficient s-stem. As we have passed to the tail of the sequence above, for any n, we may create Y', an s_n-stem such that $\operatorname{diam} Y' \leq \operatorname{diam} Y + 4\varepsilon$. But then,

$$d^+_{s_n}(x,y) \leq \operatorname{diam} Y' \leq \operatorname{diam} Y + 4\varepsilon \leq d^+_s(x,y) + 5\varepsilon.$$

After taking limits and letting $\varepsilon \to 0$, we conclude $\lim_{s_n} d^+_{s_n}(x,y) = d^+_s(x,y)$. □

Corollary 3.16. *Suppose that φ_s is a pinched Jordan curve. The inner distance*

$$\begin{aligned} d_\cdot(\cdot,\cdot) : S^1 \times S^1 \times [0,\infty] &\longrightarrow [0,\infty) \\ (x,y,s) &\longmapsto d_s(x,y) \end{aligned}$$

is continuous.

Proof. We define $d^+_\infty(x,y) = \lim_{s\to\infty} d^+_s(x,y)$ and by Proposition 3.15 this extension is well defined. Suppose $x_n \to x$ and $y_n \to y$ are convergent sequences in S^1 and $s_n \to s \in [0,\infty]$. Then

$$\begin{aligned} \left|d^+_s(x,y) - d^+_{s_n}(x_n,y_n)\right| &\leq \left|d^+_s(x,y) - \left(d^+_{s_n}(x_n,x) + d^+_{s_n}(x,y) + d^+_{s_n}(y_n,y)\right)\right| \\ &\leq \left|d^+_s(x,y) - d^+_{s_n}(x,y)\right| + d^+_{s_n}(x_n,x) + d^+_{s_n}(y_n,y). \end{aligned}$$

The first term goes to zero by Proposition 3.15 and the latter terms by Lemma 3.11. □

Proof of Theorem 3.5. Property 1 is Proposition 3.9. Continuity of d^+_s and its extension are Proposition 3.15 and Corollary 3.16; $S^1 \times S^1 \times \{\infty\}$ is nowhere dense in $S^1 \times S^1 \times [0,\infty]$, so d^+_∞ is the unique such extension. As a limit of metrics, d^+_∞ is symmetric and obeys the triangle inequality; unless C_∞ is embedded, it may be the case that $d_\infty(x,y) = 0$ for distinct $x, y \in S^1$, so d^+_∞ is a pseudo-metric. □

4. The Inner Decompositions of a Pinched Jordan Curve

In this section, we show that associated to every surjective pinched Jordan curve there is a pair of transverse singular foliations of the disk $\mathcal{D}$ extending the decompositions $\Lambda^\pm$. The following are definitions from [FL25]:

Definition 4.1. A decomposition Λ of a circle S is *special* if it is upper semi-continuous and has the following properties:

(1) *unlinking*: distinct $\lambda, \lambda' \in \Lambda$ are unlinked
(2) *nesting*: given any $\lambda \in \Lambda$ and a disjoint compact interval $I \subset S$, there is a decomposition element $\lambda' \in \Lambda$ which separates I from λ.

A pair of decompositioins $\Lambda^\pm$ of S have *efficient intersection* if $\lambda^+ \cap \lambda^-$ is at most one point for any $\lambda^+ \in \Lambda^+$ and $\lambda^- \in \Lambda^-$.

We call a pair of special decompositions $\Lambda^\pm$ of S an *especial pair* if they have the efficient intersection property. ♮

Definition 4.2. Let D denote a closed disk and let S be its boundary. A decomposition $\mathcal{S}$ of D is *unbounded* if each $K \in \mathcal{S}$ intersects S and it is *monotone* if each of its elements are connected.

A pair of decompositions $\mathcal{S}^\pm$ have *efficient intersection* if $K^+ \cap K^-$ is at most one point for each $K \in \mathcal{S}^+$ and $K^- \in \mathcal{S}^-$.

A pair of monotone, upper semi-continuous, unbounded decompositions $\mathcal{S}^\pm$ of D are called an *muu* pair. If the pair also has efficient intersection, then they form an *emuu* pair. ♮

Theorem 4.3 ([FL25], Theorem 3.4)**.** *Let $\Lambda^\pm$ be an especial pair on a circle S. Then there are decompositions $\mathcal{F}^\pm$ of a disk $\mathcal{D} = D \sqcup S$ with the following properties:*

(1) *$\mathcal{F}^{\pm}$ are an emuu pair with $\partial\mathcal{F}^{\pm} = \Lambda^{\pm}$ and is the unique pair up to a homeomorphism of $\mathcal{D}$ that fixes S.*
(2) *The decompositions $\mathcal{F}^{\pm}$ determine transverse pair of essential foliations of D given by*

$$F^{\pm} = \{\textit{Components of } K \cap D \mid K \in \mathcal{F}^{\pm}\}$$

Theorem 4.11 follows from showing that for pinched Jordan curves with no left or right side, the pinched pseudometrics d_∞ determine an especial pair $\Lambda^{\pm}$.

Definition 4.4. Let φ be a pinched Jordan curve and $d^{+}_{\infty}\pm$ corresponding the pinched pseudometrics. Define the *left inner decomposition* $\sim_+$ (resp. *right inner decomposition* $\sim_-$) on S^1 by $x \sim_+ y$ if $d^{+}_{\infty}(x, y) = 0$ (resp. $x \sim_- y$ if $d^{-}_{\infty}(x, y) = 0$). Let $\Lambda^{\pm}$ denote the set of equivalence classes of $\sim_{\pm}$ ♮

Verifying that $\sim_{\pm}$ determine an equivalence follows directly from the definition of $d^{\pm}_{\infty}$.

Lemma 4.5. *Every $\lambda \in \Lambda^{\pm}$ is closed as a subset of S^1.*

Proof. Let $\lambda \in \Lambda^{+}$ and let x_n be a sequence of points in λ converging to some point $x \in S^1$. For all n, $d^{+}_{\infty}(x_1, x_n) = 0$. As d^{+}_{∞} is continuous, it follows that $d^{+}_{\infty}(x_1, x) = 0$ as well. □

The following is essentially [Cal07, Lemma 10.13].

Lemma 4.6. *Let λ and λ' be distinct equivalence classes in Λ^{+}. Then λ and λ' are unlinked as subsets of S^1*

Proof. Assume that $\lambda, \lambda' \in \Lambda^{+}$ link in S^1. In particular, we can find $x, y \in \lambda$ and $z, w \in \lambda'$ such that $\{x, y\}$ and $\{z, w\}$ link. Let $s_n \to \infty$ and suppose X_n are taut s_n-stems for x and y and Z_n are taut s_n-stems for z and w. As $\{x, y\}$ and $\{z, w\}$ link, it must be the case that $X_n \cap Z_n \neq \varnothing$ for all n. But then $X_n \cup Z_n$ is an s_n-stem for x and z and $\operatorname{diam}(X_n \cup Z_n) \leq \operatorname{diam} X_n + \operatorname{diam} Z_n$. Thus, $d^{+}_{\infty}(x, z) = 0$ and therefore $\lambda = \lambda'$. □

Lemma 4.7. *If $x, y \in S^1$ are distinct, then at most one of $d^{\pm}_{\infty}(x, y) = 0$.*

Proof. Suppose to the contrary. Fix sequences $\varepsilon_k \to 0$ and $s_k \to \infty$ and let α_k and β_k denote ε_k-efficient stems in $\overline{C_{s_k^+}}$ and $\overline{C_{s_k^-}}$ respectively. Then $J_k = \alpha_k \cup \beta_k$ is a Jordan curve. If U is any neighborhood of $p = \varphi_\infty(x)$, we may throw away finitely many terms and assume $J_k \subset U$. Let V_k denote the component $S^2 \setminus J_k$ contained in U and I_k the component of $S^1 \setminus \{x, y\}$ such that $\varphi_k(I_k) \subset V_k$. After passing to the tail of the sequence, we may assume that I_k is the same component for all k—so we write I for I_k. We know that $\varphi_k(I) \subset V_k$ and hence $\varphi_\infty(I) \subset \{p\}$. As φ_∞ must have totally disconnected preimages, we have arrived at a contradiction. □

Lemma 4.8. *The decompositions $\Lambda^{\pm}$ are upper semi-continuous.*

Proof. Let $K \subset S^1$ be closed. We will show that $L = \bigcup\{D \subset \mathcal{D} \mid D \subset U\}$ is closed—it will then follow, by Proposition 2.9 (2), that Λ^{+} is then upper semi-continuous. Suppose that x_k is a sequence in L converging so some $x \in S^1$. Associated to each x_k is a leaf λ_k. After passing to a subsequence if necessary, assume that $\lambda_k \xrightarrow{HD} \lambda$. By construction, there $y_k \in \lambda_k \cap K$; as K is compact we may pass to another subsequence and assume that $y_k \to y \in K$. As $x_k, y_k \in \lambda_k$, $d^{+}_{\infty}(x_k, y_k)$ for all k. But d^{+}_{∞} is continuous so $d^{+}_{\infty}(x, y) = 0$ as well. Thus, $x \in L$ and L is closed. □

Let X denote $\mathbb{R}^2$ or S^2. By a *rectangle* in X we mean an embedding of $[0, 1] \times [0, 1]$ in X. We say that a closed disk $D \subset X$ is μ-thin when $\mu = \sup\{r \mid B_r(p) \subset D \text{ for some } p \in D\}$.

Lemma 4.9. *Suppose $Q \subset \mathbb{R}^2$ is a rectangle and let L, R, T, B denote the left, right, top, and bottom edged of the rectangle respectivley. Suppose further that there are constants δ, μ such that*

(1) $d_Q(L, R) > \delta$, where d_Q is inner distance on Q.
(2) Q is μ-thin

If $\mu < \delta/2$, then there are points $p \in T$, $q \in B$ such that $d_Q(p, q) < 2\mu$, $d_Q(p, L \cup R) > \delta/2$, and $d_Q(q, L \cup R) > \delta/2$.

Proof. In this proof, we use the d_Q metric to create our neighborhoods. As $d_Q(L, R) > \delta$, $N_{\delta/2}(L) \cap N_{\delta/2}(R) = \varnothing$. Let γ be any path in $Q \setminus (N_{\delta/2}(L) \cup N_{\delta/2}(R))$ starting in $T \setminus (N_{\delta/2}(L) \cup N_{\delta/2}(R))$ and ending in $B \setminus (N_{\delta/2}(L) \cup N_{\delta/2}(R))$. As Q is μ-thin, $\{N_\mu(L), N_\mu(R), N_\mu(T), N_\mu(B)\}$ covers Q, but as $\mu < \delta/2$ so does $\{N_{\delta/2}(L), N_{\delta/2}(R), N_\mu(T), N_\mu(B)\}$. As $\gamma \subset Q \setminus (N_{\delta/2}(L) \cup N_{\delta/2}(R))$, $\{N_\mu(T), N_\mu(B)\}$ covers γ. It follows that $N_\mu(T) \cap N_\mu(B) \cap \gamma$ is nonempty—indeed $N_\mu(T) \cap \gamma$ and $N_\mu(B) \cap \gamma$ are open subsets of γ covering γ, hence they must intersect as γ is connected.

Thus, there is some point $x \in \gamma \cap N_\mu(T) \cap N_\mu(B)$. By construction, there are point $p \in T$, $q \in B$ such that $d_Q(x, p), d_Q(x, q) \leq \mu$—thus $d_Q(p, q) \leq 2\mu$, as required. □

Proposition 4.10. *Let φ be a pinched Jordan Curve with no left side. Then Λ^+ satisfies the nesting condition.*

Proof. Select $\lambda \in \Lambda^+$, $U = (a, b)$ a complementary region to λ, and let $I = [x_1, x_2] \subset U$ be a closed interval. Since I is contained in a complementary interval to λ, the images $\varphi(I)$ and $\varphi(\lambda)$ must be disjoint. Let $\delta = \inf_{x \in I} d^+_\infty(a, x)$; as d^+_∞ is continuous, $\delta > 0$.

We now fix some $x \in I$ and choose a sequence $s_n \to \infty$. For each n, let α_n be a $1/n$-efficient s_n-stem for x and a. Observe that for each n, there is a rectangle $Q_n \subset \overline{C^+_{s_n}}$ with sides α_n, $\varphi_{s_n}(I)$, and $\varphi_{s_n}(U \setminus I)$—denote these as L_n, R_n, T_n, B_n respectively. First, we will find points z, w in different comopnents of $U \setminus I$, then we will show that (z, w) is a leaf in Λ^+.

For each n, let ε_n denote the thinnness of $\overline{C^+_{s_n}}$; note that $\varepsilon_n \to 0$. As φ_∞ has no left side, if n is sufficiently large, then we may assume that $\varepsilon_n < \delta/4$ and $d_{s_n}(L_n, R_n) > \delta/2$. These are the exact hypotheses of Lemma 4.9, so there are points $p_n \in T_n$ and $q_n \in B_n$ such that $d_{Q_n}(p_n, q_n) < 2\varepsilon_n$. Now let $z_n, w_n \in U \setminus I$ be sequences such that $\varphi_{s_n}(z_n) = p_n$ and $\varphi_{s_n}(w_n) = q_n$. Passing to a subsequence if necessary, assume that $z_n \to z, w_n \to w$. As every p_n and q_n are at least $\delta/4$ away from L_n and R_n, $p, q \in U \setminus I$ as well.

Then

$$d_{s_n}(z, w) \leq d_{s_n}(z, z_n) + d_{s_n}(z_n, w_n) + d_{s_n}(w_n, w)$$

The outer terms tend to zero by Lemma 3.11 and the central term is bounded by $2\varepsilon_n$ by construction. Taking limits, we have some leaf λ' containing z and w. As z and w lie in different components of $U \setminus I$, λ' separates λ and I. □

Decompositions of S^1 with the nesting property have a rigid structure. Suppose that $\{x, y\} \in \Lambda^+$ contains exactly two points and write $S^1 \setminus \{x, y\} = I \sqcup J$. There are sequences of leaves $\lambda_n \subset I$ and $\mu_n \subset J$ such that $\lambda_n \xrightarrow{HD} \{x, y\}$ and $\mu_n \xrightarrow{HD} \{x, y\}$. On the other hand, if $\lambda \in \Lambda^+$ contains more than two points, then it determines a (possibly infinite sided) polygon inscribed in S^1 and every side of this polygon can be realized as a limit of leaves contained in the corrosponding component of $S^1 \setminus \lambda$.

Theorem 4.11. *Let φ be a surjective pinched Jordan curve and $\mathcal{D} = D \sqcup S^1$. Associated to φ is a pair of laminations $\Lambda^\pm$ which extend to pair of transverse singular foliations $F^\pm$ of D extending $\Lambda^\pm$.*

Proof. By Lemmas 4.5 to 4.8 and proposition 4.10, $\Lambda^{\pm}$ form an especial pair in S^1. Then Theorem 4.3 produces a pair of decompositions $\mathcal{F}^{\pm}$ of $\mathcal{D}$ extending $\Lambda^{\pm}$ and $F^{\pm} = \{\text{Components of } K \cap D \mid K \in \mathcal{F}^{\pm}\}$ produces the required singular foliations. □

5. Special Decompositions are Dendrites

In this section we study the decomposition spaces $\Lambda^{\pm}$ of a pinched Jordan curve. In particular, we show that they are each homeomorphic to a compact *real tree*, a uniquely arcwise connected and locally arcwise connected space; the prototypical example of a real tree is an $\mathbb{R}$-tree considered without its metric. On the other hand, if real tree T is metrizable, by [MO90] and [Bow99] T can be given a metric where it is an $\mathbb{R}$-tree. A *dendrite* is a compact, separable real tree. Given any separable $\mathbb{R}$-tree T, we may give T the weak *observer's topology* and then compactify T into a dendrite [CHL07].

Definition 5.1. A decomposition $\mathcal{D}$ of a space X is said to be *nonseparated* if, given any $D \in \mathcal{D}$, no $D' \in D$ lies in more than one component of $X \setminus D$. ♮

Definition 5.2. A decomposition $\mathcal{D}$ of a space X is said to be *saturated* if, given any $D \in \mathcal{D}$ and $x \in X \setminus D$, there is some $D' \in \mathcal{D}$ separating x and D. ♮

In [Why63], it is first shown that any nonseparated saturated decomposition of a compact space is automatically upper semicontinuous (VII.4 Thm. 4.21). If the initial space is a continuum, there is more to say.

Proposition 5.3 ([Why63], VII.4 Thm. 4.23)**.** *Let $\mathcal{D}$ be a nonseparated and saturated decomposition of a continuum X. Then the decomposition space $\mathcal{D}$ is a dendrite.*

We will show that $\Lambda^{\pm}$ are nonseparated and saturated. To do so, we will work both $\Lambda^{\pm}$ and a corresponding muu decomposition $\mathcal{F}^{\pm}$ of $\boldsymbol{\mathcal{D}} = D \sqcup S^1$ extending $\Lambda^{\pm}$. We justify this with the following lemma.

Lemma 5.4. *Let Λ be a special decomposition of S^1 and $\mathcal{F}$ a muu decomposition which extends Λ. As decomposition spaces, Λ and $\mathcal{F}$ are homeomorphic*

Proof. Consider the inclusion $S^1 \hookrightarrow \boldsymbol{\mathcal{D}}$. Since $\mathcal{F}$ extends Λ, the inclusion respects the decompositions Λ and $\mathcal{F}$ and therefore descends to a continuous bijection between Λ and $\mathcal{F}$. Λ is compact and by Proposition 2.9 $\mathcal{F}$ is Hausdorff, hence they are homeomorphic. □

We state the theorem below for a special decomposition of the circle, but we note that the proof automatically extends the result to any muu decomposition of a closed disk as well.

Theorem 5.5. *Let Λ be a special decomposition of S^1. Then the decomposition space Λ is a dendrite.*

Proof. By Lemma 5.4, we may use any muu decomposition $\mathcal{F}$ that extends Λ, so we will assume that $\mathcal{F}$ is the hull decomposition $\mathcal{F} = H(\Lambda)$. It is clear that monotone decomposition are nonseparated, so we must establish that $\mathcal{F}$ is saturated. Fix $K \in \mathcal{F}$ and $x \in \boldsymbol{\mathcal{D}} \setminus K$. Let $L = L(x) \in \mathcal{F}$ denote the unique decomposition element containing x, L_Λ the restriction of L to Λ and K_Λ the restriction of K to Λ. Explicitly, L_Λ and K_Λ are given by intersection L and K with S^1. As distinct elements of Λ are unlinked, some complementary region U of $S^1 \setminus K_\Lambda$ contains all of L_Λ. Let $I \subset U$ denote the convex hull of L_Λ. Then I is a closed interval in the complement of K_Λ, so nesting implies there is a some $C \in \Lambda$ separating K_Λ and I—hence separating K_Λ and L_Λ. As $\mathcal{F}$ is the hull decomposition, it follows that $H(C)$ separates K and L. Thus, $\mathcal{F}$ is a dendrite by Proposition 5.3 and by Lemma 5.4 so is Λ. □

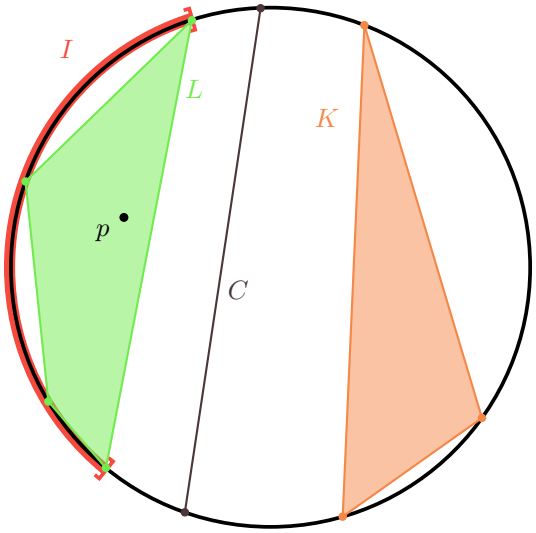


FIGURE 1. TODO

6. Self Bumping and Laminations

In [CL26], Calegari and Loukidou discus the notion of a "self-bumping" $\varphi \in \operatorname{Fr}\operatorname{Emb}(S^1, S^2)$. Such a φ can be approximated by embeddings in more than one way—the pinched laminations give a partial answer to untangling the ambiguity of self-bumping points.

We say that two maps $f, g : X \to Y$ between metric spaces are δ-*isotopic* if there is an isotopy $F : X \times I \to Y$ between f and g such that for all $x \in X$, $\operatorname{diam}\{x \times I\} < \delta$. Define an equivalence relation on the set of pseudo-isotopies landing on a fixed $\xi \in \operatorname{Fr}\operatorname{Emb}(S^1, S^2)$ as follows: $\varphi \sim \psi$ if for every $\varepsilon > 0$ there is a $\delta > 0$ with $\delta \to 0$ as $\varepsilon \to 0$ such that if for all $x \in S^1$, $d(\varphi_s(x), \xi(x)) < \varepsilon$ and $d(\psi_s(x), \xi(x)) < \varepsilon$ then φ_s and ψ_s are δ-isotopic. The goal of this final section is to prove the following:

Theorem 6.1 (Calegari-Kerbs)**.** *For a fixed $\xi \in \operatorname{Fr}\operatorname{Emb}(S^1, S^2)$ and two pseudo-isotopies Φ and Ψ landing on ξ, $\Phi \sim \Psi$ if and only if the pinched laminations agree.*

Appendix A. Cannon-Thurston Maps and Pseudo-Isotopies

In this section we show that the Cannon-Thurston maps arising from surface bundles [CT07] and from quasigeodesic flows [Fra15] are approximated by pseudo-isotopy. As a consequence, the corrosponding singular foliations agree with those constructed in Theorem 4.11.

A.1. Shrinkable Decompositions.

Definition A.1. Let X be a compact metric space, $\mathcal{D}$ an upper semi-continuous decomposition of X, and ρ any metric on $\mathcal{D}$. We say $\mathcal{D}$ is *shrinkable* if, for every $\varepsilon > 0$ there exists a homeomorphism $h : X \to \mathcal{D}$ such that:

(1) $\rho(\pi h(x), \pi(x)) < \varepsilon$ for all $x \in X$
(2) $\operatorname{diam} h(D) < \varepsilon$ for all $D \in \mathcal{D}$. ♮

One can make sense of shrinkable decomposition in the non-metric case. The reformulation involves comparing arbitrary open covers to arbitrary saturated open covers. The definition

says that elements of $\mathcal{D}$ can be made small without moving points too far relative to $\mathcal{D}$. This can be upgraded with a routine application of the Lebesgue covering lemma.

Proposition A.2 ([Dav], II.5 Thm. 2)**.** *Let $\mathcal{D}$ be an upper semi-continuous decomposition of a compact metric space X. The decomposition map $\pi : X \to \mathcal{D}$ can be approximated by homeomorphisms if and only if $\mathcal{D}$ is shrinkable.*

Remark A.3. Shrinkable decompositions preserve local topological properties. Suppose that that X is a compact metric space and $\mathcal{P}$ a topological property which applies to subsets of a space. Suppose that every $x \in X$ has arbitrarily small neighborhoods with property $\mathcal{P}$. Then every element D of the decomposition space $\mathcal{D}$ has arbitrarily small neighborhoods with property $\mathcal{P}$. ♮

Given a compact set $K \subset X$, we say that K is *cell-like* if, for every neighborhood U of K, the inclusion $\imath : K \hookrightarrow U$ is nullhomotopic. In the case where X is a 2-manifold, every cell-like set is cellular—that is, it can be written as the intersection of countably many nested closed balls. Every cellular set is evidently cell-like, but the converse is false—counterexamples can be found in [Dav] or [Kal21]. A decomposition $\mathcal{D}$ is said to be cell-like if every element is cell-like.

Lemma A.4. *Every compact, contractable set K in a manifold X is cell-like.*

Lemma A.5 ([Kal21] Thm. 5.4, (1))**.** *Let $\mathcal{D}$ be an upper semi-continuous cell-like decomposition of a surface X. Then $\mathcal{D}$ is shrinkable.*

In the context of Proposition A.2, one might wonder if the sequence of homeomorphisms approximating π can be interpolated to a pseudo-isotopy.

Definition A.6. If X has an upper semi-continuous decomposition $\mathcal{D}$ such that $\mathcal{D} = \{\varphi_\infty^{-1}(y) \mid y \in Y\} \setminus \varnothing$, then we say that $\mathcal{D}$ is *realized by pseudo-isotopy.* ♮

The exact decompositions which admit realization by pseudo-isotopy have been well studied. For our work, the following suffices.

Proposition A.7 ([Dav], II.13 Thm. 4)**.** *If $\mathcal{D}$ is a shrinkable decomposition of a manifold M then $\mathcal{D}$ can be realized by a pseudo-isotopy. Further, if $\mathcal{U}$ is a $\mathcal{D}$-saturated open cover of M, then there exists a pseudo-isotopy $\varphi : M \times I \to M$ such that for each $p \in M$, $\varphi(\{p\} \times I) \subset U$ for some $U \in \mathcal{U}$.*

A.2. **Cannon-Thurston maps are realized by pseudo-isotopy.** Given a quasigeodesic flow Ψ on a closed hyperbolic 3-manifold M, Frankel constructs a $\pi_1 M$ equivariant Peano curve $S^1 \to S^2$ [Fra15]. Here, the S^1 is the compactified boundary of the flowspace of Ψ and the S^2 is the sphere at infinity of $\tilde{M}$. This generalizes the earlier work of Cannon and Thurston [CT07] who construct a Peano curve in the case where M fibers over the circle. These Peano curves are all realizable by pseudo-isotopy. By Proposition A.7, it is sufficient to construct a shrinkable decomposition such that the decomposition map realizes the Peano curve.

If M fibers over the circle, then M comes equipped with a suspension pseudo-Anosov flow Φ given by the monodromy. Let $\boldsymbol{\mathcal{P}} = \mathcal{P} \sqcup S^1_u$ denote the compactified flowspace of Φ (which we may view as the universal cover of some fiber Σ). Glue two copies of $\boldsymbol{\mathcal{P}}$ along S^1_u to form the *universal sphere*, S^2_u. $\boldsymbol{\mathcal{P}}$ supports a pair of tranverse singular foliations—each of which is a monotone, upper semi-continuous decomposition. We may extend these foliations to a decomposition $\mathcal{D}_\Phi$ of S^2_u by foliating the northern and southern hemisphere with the

positive and negative foliations respectively. We stress that S^2_u is not foliated—points in S^1_u generally lack foliated charts. As no positive leaf shares an end with a negative leaf, no leaf of $\mathcal{D}$ can separate S^2_u.

Notice that there is a natural map $\pi : S^2_u \to S^2_\infty$ given by the Cannon-Thurston map on S^1_u and collapsing each decomposition element in the hemispheres to the image of their corresponding point in S^2_∞. Then $\mathcal{D}_\Phi = \{\pi^{-1}(p) \mid p \in S^2_\infty\}$, so $\mathcal{D}$ is upper semicontinuous. As each decomposition element is cellular (indeed, it is either an arc, a finite pronged singular arc, or a point), $\mathcal{D}$ is cellular, hence by Lemma A.5 $\mathcal{D}$ is shrinkable and therefore realized by a pseudo-isotopy.

The case where Ψ is a quasigeodesic flow on a closed hyperbolic manifolds is similar. The flowspace $\boldsymbol{\mathcal{P}} = \mathcal{P} \sqcup S^1_u$ comes equipped with a pair of decompositions $\mathcal{D}^\pm$. We may construct the universal sphere the same way: $S^2_u = \boldsymbol{\mathcal{P}} \cup_{S^1_u} \boldsymbol{\mathcal{P}}$ and we can use $\mathcal{D}^\pm$ to generate a decomposition of S^2_u.

There are natural maps $e^\pm : \mathcal{P} \to S^2_\infty$ sending a flowline α to its forward or backward endpoint. In [Fra15], Frankel shows that that $e^\pm$ have natural extensions to S^1_u. Consider the map $m : S^2_u \to S^2_\infty$ which is e^+ on the (closed) northern hemisphere and e^- on the southern. As e^+ and e^- agree on S^1_u, m is well defined. In [Fra17, Section 5], Frankel uses m to construct the *master decomposition* of S^2_u:

$$\mathcal{M} = \{m^{-1}(p) \mid p \in S^2_\infty\},$$

a monotone and upper semi-continuous decomposition of S^2_u. Notice that this is the decomposition given by taking the transitive closure of $\mathcal{D}^\pm$ viewed as partial decomposition of S^2_u. He then shows that each $K \in \mathcal{M}$ is nonseparating—meaning $\mathcal{M}$ is a cell-like decomposition by Lemma A.4 and therefore shrinkable and realized by a pseudo-isotopy.

It is worth commenting on the pseudo-isotopy produced by Proposition A.7. We construct it by interpolating between a sequence of homeomorphisms $S^2_u \to S^2_\infty$ which shrink decomposition elements to have diameter less than some $\varepsilon_n \to 0$—see Figure 2. In both cases, the intersection of each $D \in \mathcal{D}$ with its respective S^1_u is totally disconnected, it is not shrunk and instead winds its away around S^2_∞. In the limit, of course, the image of S^1_u is all of S^2_∞, realizing the Cannon-Thurston maps.

Consider one decomposition element $K \in \mathcal{M}$ and write $\mathcal{K} = \mathcal{K}^+ \cup \mathcal{K}^-$ for its intersections with $\mathcal{D}^\pm$—note that if both $\mathcal{K}^+$ and $\mathcal{K}^-$ are nonempty, then $\mathcal{K}^+ \cap \mathcal{K}^- \subset S^1_u$. Let K^+ be a component of $\mathcal{K}^+$ and K^- a component of $\mathcal{K}^-$. Write $\{x_n\} = K^+ \cap S^1_u$ and $\{y_n\} = K^- \cap S^1_u$. Note that $\pi(\{x_n\}) = \pi(\{y_n\}) = p$, but it is not necessarily the case that $\pi^{-1}(p) = \{x_n\} \cup \{y_n\}$. Under the pseudo-isotopy ψ_s, K^+ becomes a closed, connected set contained entirely to the left of the Jordan curve $\psi_s(S^1_u)$ except at $\psi_s(\{x_n\})$, where it intersects $\psi_s(S^1_u)$—and the same can be said for $\psi_s(K^-)$'s containment to the right of $\psi_s(S^1_u)$.

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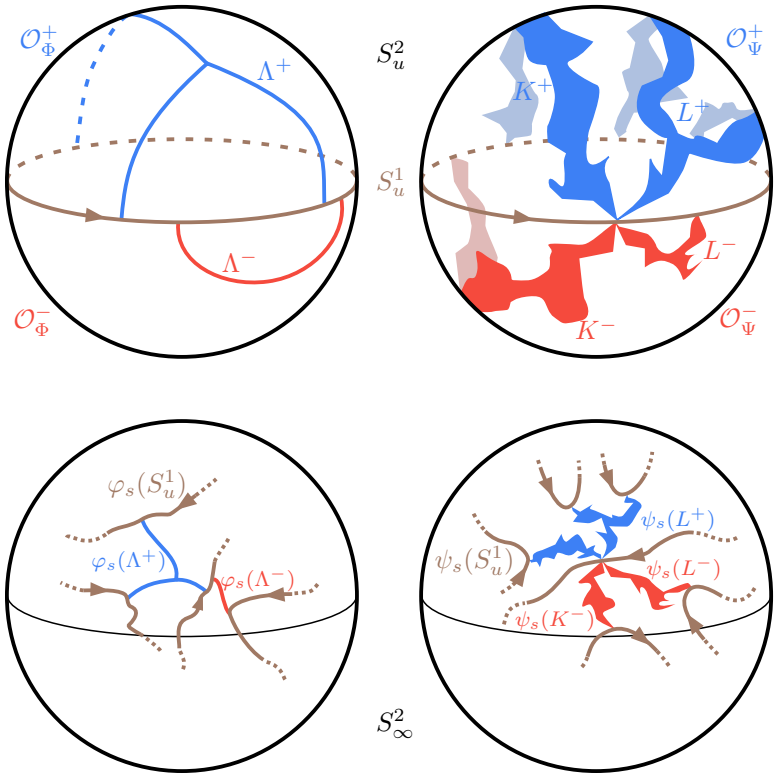


FIGURE 2. Top: The universal sphere associated a pseudo-Anosov suspension flow Φ and a quasigeodesic flow Ψ and some positive an negative decomposition elements. For a suspension flow, positive and negative decomposition elements do not meet at S^1_u, but this can happen for a quasigeodesic flow. Bottom: The image of the decomposition elements under a pseudo-isotopy realizing the decomposition.

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